CURRICULUM VITAE

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Overview

I am an experienced data analyst, having worked for four years in a consulting advisory firm, with blue-chip clients in Australia and overseas. In that role I developed and applied more advanced database automation and data presentation techniques that assisted this business in providing advice that was well-accepted by its clients.

Leveraging this experience, I completed a science degree in which I have focused on extending my expertise in programming and data analysis. As part of this degree (and through my own self-study) I have developed skills and interest in real-time graphics development including real-time physics simulation and low-level graphics rendering and computational geometry while rounding out my understanding of best practices, IT development and operational processes and honing my database, analysis and programming skills.

Work Experience

2017 – 2021: Energy Market Consulting Associates, Data Analyst / Senior Data Analyst

Role

Data analyst supporting a team of professionals in energy economics, electricity and gas sector engineering and finance. The main clients of this consulting firm comprise electricity and gas distribution and transmission businesses and energy regulators in Australia, New Zealand and Malaysia.

At the commencement this work involved Excel spreadsheeting. I introduced more advanced data analysis techniques including database automation (e.g. using VB, Python and SQL) and enhanced modelling workflow and presentation of results (e.g. using Tableau) for the company's reports to clients.

Example Projects

Standardised Excel models for electricity businesses

- **Client:** Australian Energy Regulator (part of the ACCC)
- Created standardised model templates for use by Australian electricity distribution businesses, to provide their proposed metering expenditure and proposed metering tariffs, for regulatory approval
- The models include macros allowing configuration to facilitate the requirements of specific users and a goal-seek to determine metering tariffs according to the regulator's approved methodology
- Also developed a handbook for users and led stakeholder feedback sessions at which the draft models were tested with users
- The two models and associated handbook are published on the Australian Energy Regulator's website, available here.

  https://www.aer.gov.au/industry/registers/resources/models/standardised-models-regulated-metering-services

Modelling of trials of customer response to smart metering tariffs in Malacca (Malaysia)

- Data analyst as part of a consultant team engaged by the Malaysian electricity company (Tenaga Nasional Berhad) to undertake trials of innovative tariffs based on using Advanced Metering Infrastructure that had been deployed in the state of Malacca

- Developed large-scale databases with annual half-hourly consumption of 1,000 customers in the trial
- In conjunction with mathematics professors from a Malaysian university, developed statistical methods for analysing the customer behavioral impact of different tariff structures and correlating this with the results of surveys of the sample customers.
- The results of the analysis informed the electricity company's decisions considering its future tariff strategies.

Regulatory Tariff Submission – Malaysian electricity business

- Responsible for collating forecast expenditure information for the next three years from the five divisions of the Malaysian electricity business (Tenaga Nasional Berhad). The information was collated into database form for use in assessing the implications of the proposed expenditure for the businesses' tariffs and profitability.
- Analysis included assessing forecast expenditure trends relative to historical expenditure, assessing changes in productivity, international benchmarking to assess efficiency and assistance with economic modelling of business case justifications.
- Work included participating in a major workshop in Malaysia interacting with up 70 executives from the five Malaysian electricity business entities to support the development of their budget forecasts and supporting justification.

Regulatory Tariff Submission – Malaysian gas business

- Responsible for collating forecast expenditure information for the next three years for the main Malaysian gas distributor (Gas Malaysia Berhad (GMB). The information was collated into database form for use in assessing the implications of the proposed expenditure.
- Analysis included assessing forecast trends relative to historical expenditure, international benchmarking to assess efficiency and assistance with economic modelling of business case justifications.
- The role included working in Malaysia with the gas business personnel and participated in a presentation of findings to C-suite executives in the business.

Statistical and business case analysis – Australian electricity regulatory determinations

- Created reconciled accounting forecasts and proposed comparative expenditure forecasts correlated with engineering trends and drivers using bottom up and top-down methodologies to assess the validity of proposed financial forecasts
- Reviewed business cases for major electricity expenditure proposals.
- Provided analysis to support insights, justification, audits and analysis of company projects and expenditure to inform regulatory decisions on tariffs and prices.

Regulatory benchmarking analysis

- Developed and applied techniques to scrape detailed reported information from the website of the Australian Energy Regulator, and developed a database from this information (combined with data from ABS via API and other sources) to facilitate performance benchmarking analysis between the 16 electricity distribution businesses in the Australian electricity market
- Undertook comparative benchmarking using normalization techniques and statistical correlation tests in order to draw conclusions on comparative performance and drivers for differences between the businesses
- The results of this analysis were used in advice provided to the Australian Energy Regulator on the reasonableness of forecast expenditure requirements, for the purpose of setting electricity tariffs.

Project costing for consortium – China and Australia

- Developed project costing for a consortium comprising Australian, US and Chinese companies, to jointly bid for a project in Hangzhou, China with a value of the order of $AUD15 million.
- The costing analysis needed to account for the different resources, rates and currencies of the consortium parties, structured in the format required by the Chinese purchasing entity, and involved travelling to Hangzhou to work with the consortium members.

Skills

Category	Skills
Programming Languages	C++, C#, GLSL, Javascript & WebGL, Python, VBA, Bash
Game Engines	Unity, Unreal Engine
Build & Tooling	Git, CMake, Vite, Webpack
Data Handling	Web Scraping (Python), SQL, Data Flows (Tableau)
Data Analysis & Visualization	Python, SQL, Tableau, Office Suite (Excel, Power Query)
Machine Learning	LORA & Embedding training, Diffusion based mesh generation (text->3d)
Simulation	Position-Based Dynamics (PBDS), Aerodynamics, Realtime Computational Geometry and Meshing
Game Dev	Performant networked multiplayer backends, real-time physics, PID Controllers for physics-based AI

Education

Northern Beaches Secondary College – Manly Selective Campus

University of Technology Sydney - Bachelor of Science in Game Development (2022 – 2025)
Graduated BSc with Distinction
Dean's List – Faculty of Engineering and Information Technology

Indicative Courses Completed

Course	Result
Communication for IT Professionals	High Distinction
Introduction to Information Systems	High Distinction
Programming Fundamentals	High Distinction
Web Systems	Distinction
Applications Programming	High Distinction
Business Requirements Modelling	Distinction
Fundamentals of Interaction Design	High Distinction
Introduction to Computer Game Development	Distinction
Computer Graphics	High Distinction
Data Structures and Algorithms	High Distinction
Database Fundamentals	High Distinction
Network Fundamentals	Distinction
Change-making for social impact	Distinction
Directed Study 1 (Custom Rendering Engine, Geometry Graph, Physics, ECS)	High Distinction
Game Design Studio 1	High Distinction
Interactive 3D Animation	Distinction
Programming on the Internet	High Distinction
Game Design Studio 2	High Distinction

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